

The empirical recurrence for OEIS A279922, recovered from its tail

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Abstract

OEIS A279922 records “Empirical recurrence of order 54” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 2323$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 59$. Its last coefficient is $c_{54} = -231928233984000 \neq 0$, so no further tail satisfies anything shorter; any order-54 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A279922 is “Number of $n \times 3$ 0..2 arrays with no element equal to a strict majority of its horizontal, vertical and antidiagonal neighbors, with the exception of exactly two elements, and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

0, 15, 474, 14281, 399487, 11023893, 295670293, 7789959515, ...

The entry states:

Empirical recurrence of order 54 (see link above)

As of the “Last modified” line on the live entry (Dec 23 2016 08:18:30, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 2323$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2323$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 54: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 54.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 4646$ exact terms from $n = 5$ on, it returns order 54 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{54} c_i a(n-i),$$

c_1	66	c_{15}	5307527984	c_{29}	3894984577074624	c_{43}	1051917630712578048
c_2	−1431	c_{16}	−34641778776	c_{30}	7859518539522624	c_{44}	1065737398967599104
c_3	9652	c_{17}	−200050036752	c_{31}	8342527704429696	c_{45}	97559367475265536
c_4	13839	c_{18}	−149851856896	c_{32}	−3624503683625856	c_{46}	−591896479820414976
c_5	−63804	c_{19}	124550369712	c_{33}	−43851035755441408	c_{47}	−414802109704175616
c_6	465127	c_{20}	643491921744	c_{34}	−106605201147806976	c_{48}	31781517281722368
c_7	−1224912	c_{21}	6791334336448	c_{35}	−98883274129807872	c_{49}	153134522626473984
c_8	−8708514	c_{22}	16988881599312	c_{36}	118908400711294464	c_{50}	39856770373386240
c_9	5629902	c_{23}	4202064489888	c_{37}	478401153847209984	c_{51}	−19143451996061696
c_{10}	−24112776	c_{24}	−41393977860192	c_{38}	536897327219159040	c_{52}	−7050811586641920
c_{11}	−48326316	c_{25}	−149162773829088	c_{39}	−98135012702560256	c_{53}	2794735219507200
c_{12}	1038927416	c_{26}	−439563469654656	c_{40}	−1016474881168146432	c_{54}	−231928233984000
c_{13}	1621459428	c_{27}	−588363955976576	c_{41}	−1152470793758244864		
c_{14}	−753521556	c_{28}	605956032360960	c_{42}	−112742532187029504		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 59$, and it is the recurrence OEIS A279922 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{54} - c_1 x^{54-1} - \dots - c_{54}$. Its constant term is $-c_{54} = 231928233984000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 54 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 54, so $\deg q = 54$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 54, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 54, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -231928233984000 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-54 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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